

Master thesis presentation

Good afternoon everyone. My thesis is titled "Investigations in some models in gradient elastoplasticity", supervised by professor Rindler from the University of Warwick and professor Giusti from the University of Padova.

*and explain what are gradient ^{elasto}plasticity models.

Before presenting the models addressed in the thesis I would like to briefly mention the mechanical phenomenon underlying them. *It is the phenomenon of the localization of the plastic distortions in thin regions of an elastoplastic materials. The two concrete examples that motivated the investigations of the thesis,

are the necking of a metal bar in the tensile test that is the localized narrowing of the cross section of the bar that is observed when it is subject to elongation, and the formation of shear bands in geomaterials, that are thin bands where shear deformations concentrates. From the modelling viewpoint

In the classical theory of elastoplasticity theory the plastic distortion evolves independently on each point of the material so that the plastic distortion is free to localize in hypersurfaces leading to discontinuous displacements. Thus the associated boundary value problem cannot be well-posed in Sobolev spaces. In gradient elastoplasticity sharp variations of the plastic distortion are penalized including a dependence of the free energy of the system on the gradient of the plastic distortion, restoring the existence of solutions in Sobolev spaces.

* associated with the two previously mentioned examples

In my thesis I developed two gradient elastoplasticity models * and I studied the existence of solutions to the related boundary value problems using the tools of nonlinear PDE theory and calculus of variations. One of them is a simplified one-dimensional elastoplastic model of a metal bar which I used to perform a stability analysis with the final aim of studying the necking phenomenon. The other model is a nonclassical approach to the modeling of geomaterials, or more in general of materials that allows for plastic dilatancy and pressure sensitive yield strength. In this case the aim was to overcome the difficulties implied by non-associated flow rules that are commonly employed in this context. In this presentation I will focus exclusively to this second model for which I obtained more complete results.

I first need to explain the general theoretical framework that I exploited to formulate the model. According to this framework, we view the configurations of a mechanical system as the elements z of a Banach space $\mathcal{Z}$. The generalized forces acting on the system are viewed as the elements of the dual space $\mathcal{Z}^*$. The equilibrium of the system is formulated as the balance between the internal force π and the external force x and must hold at every time

of the evolution. The external force is assigned as a datum of the problem while the internal force is linked to the kinematics through constitutive relations. Specifically we choose a free energy Ψ depending on the configuration and a dissipation potential Φ depending on the rate of the configuration. Then the internal force is written as the sum of the differential of the free energy, that gives the conservative contribution, and the differential of the dissipation potential that gives the dissipative contribution. The dissipation potential is not assumed to be smooth so that we use the generalized notion of subdifferential that is a multivalued map. This is the reason why the constitutive relation is written as an inclusion rather than an equality. Now the evolution problem can be formulated as this inclusion that is called doubly nonlinear equation, together with an initial condition.